VIP: Interacting with Sound and Space

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In dedicated projects and teams, L42i's VIP explores how we can interact with spatial audio - in artistic and scientific contexts. Many of our projects use 3D multichannel loudspeaker systems, running on Linux Audio servers. Our work combines design and engineering with creative applications and methods from Human Computer Interaction research. By combining concerts, installations, and experiments we have a holistic perspective, ideal for interdisciplinary teams. **MEETING TIMES:** Individual for each project team

BIKES

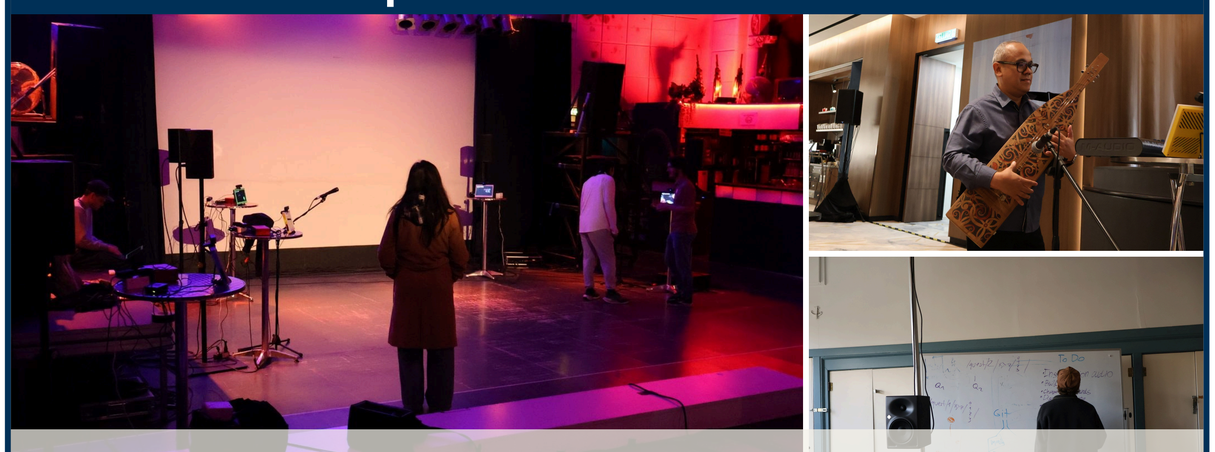


BIKES combines industrial design with all kinds of engineering. The four networked bikes with loudspeakers can be used for critical interventions in the urban space. Our latest version runs AI-supported camera tracking. Next step: new hardware design (the black edition).

MAJORS NEEDED

Industrial Design, Music Tech, Computer Science, Electrical Engineering, Public Policy, Literature, Media, and Communication, ...

MËSH and SuperMËSH



This project explores distributed systems for immersive multi-user interaction. MËSH has traveled a lot - across the US, to Europe and Asia - to create experiences for musicians and audiences. With SuperMËSH we are taking it to a next level with distributed multi-camera movement tracking.

MAJORS NEEDED

Computer Science, Computational Media, Electrical Engineering, Music Tech, ...

XR Music Interaction



Our lab explores how we can combine XR interfaces with 3D loudspeaker systems for immersive interactive experiences. This lets musicians and everyone move sound in space through augmented objects. We explore new methods for evaluating the interaction in creative contexts.

MAJORS NEEDED

Music Tech, Computer Science, **Computational Media**, Electrical Engineering, Industrial Design, ...

brain.wav



In collaboration with TReNDS we are exploring how we can use immersive loudspeaker systems to make fMRI data audible. This HCI research aims at tools for neuroscience researchers. With real-time input devices, they can navigate the recorded data, while hearing it.

MAJORS NEEDED

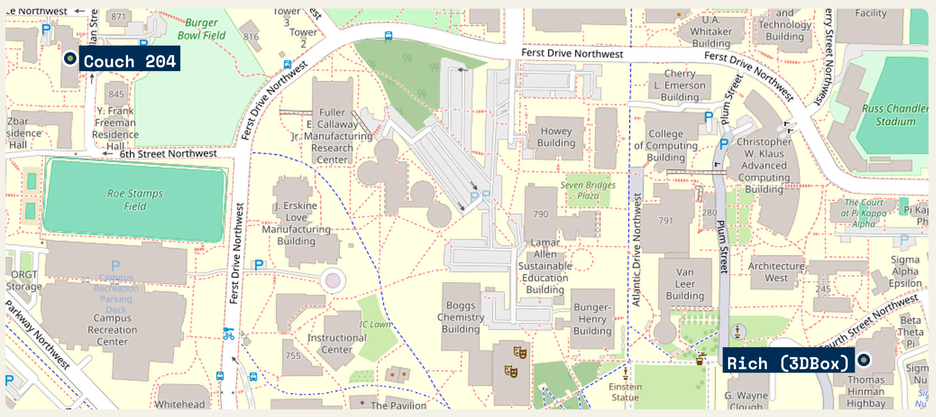
Music Tech, Computer Science, **Neuroscience**, Biomedical Engineering, Psychology, ...

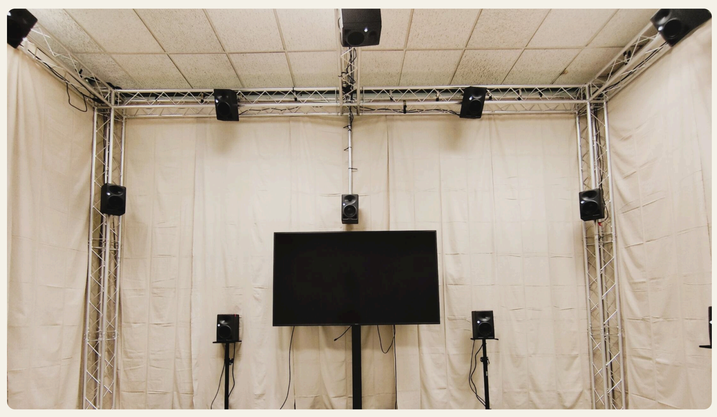
OUR SPACES



Couch 204

Main multi-purpose lab space with a 16-channel dome, desks and project areas.





Rich (3DBox)

Immersive sound system with 29 channels for research and artistic practice.